

Toy model for migrating magnetic flux patches across the core-mantle boundary

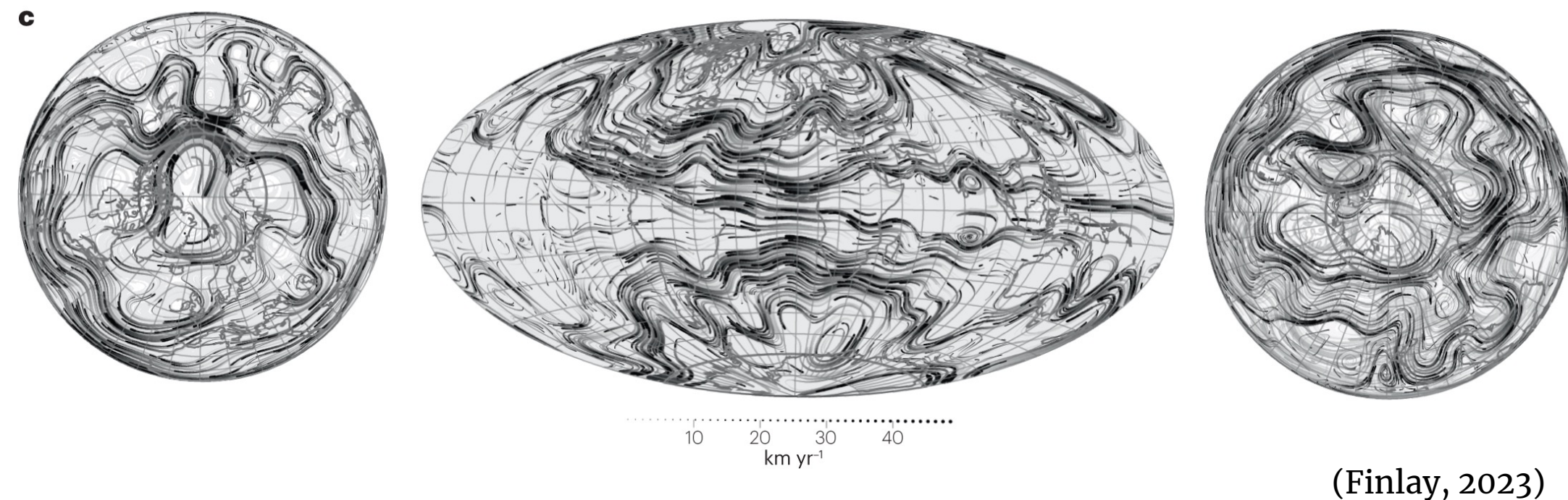
Nicole Clizzie & Catherine Constable

Institute of Geophysics & Planetary Physics, Scripps Institution of Oceanography,
University of California San Diego, La Jolla, CA 92093-0225, USA

Motivation & Background

Assess whether the planetary gyre is a persistent feature in the magnetic field that can be detected with a limited number of paleomagnetic field records from sediment cores and archeological artifacts

Core surface flow



(Finlay, 2023)

Earth's axial dipole moment (ADM) variations

- Proposed mechanism for growth and decay is meridional flux advection (Finlay, 2016)
- Magnetic flux patches migrating poleward or equatorward
- Levantine Iron Age Anomaly (~3500 – 500 BP)
 - Strong regional increase in paleofield intensity
 - Possible spike at the core-mantle boundary (CMB)
 - Spike migrates northwest, increasing ADM
- Planetary-scale gyre
 - Anti-cyclonic motion in the Atlantic hemisphere
 - Contributing to the decay of ADM in southern hemisphere
 - Transports normal polarity magnetic flux toward the equator
 - Transports reverse polarity magnetic flux (South Atlantic Anomaly) towards the south pole
 - ~700–1,100 year cycle of gyre (frozen flux approximation of core surface flow)

Method: Toy Model

Spherical triangle tessellation (STT) spatial basis

- Create a spike at the CMB in a dipolar background field
- Model migrating spike in time
- Convert to spherical harmonics
- Calculate ADM, field directions and strength

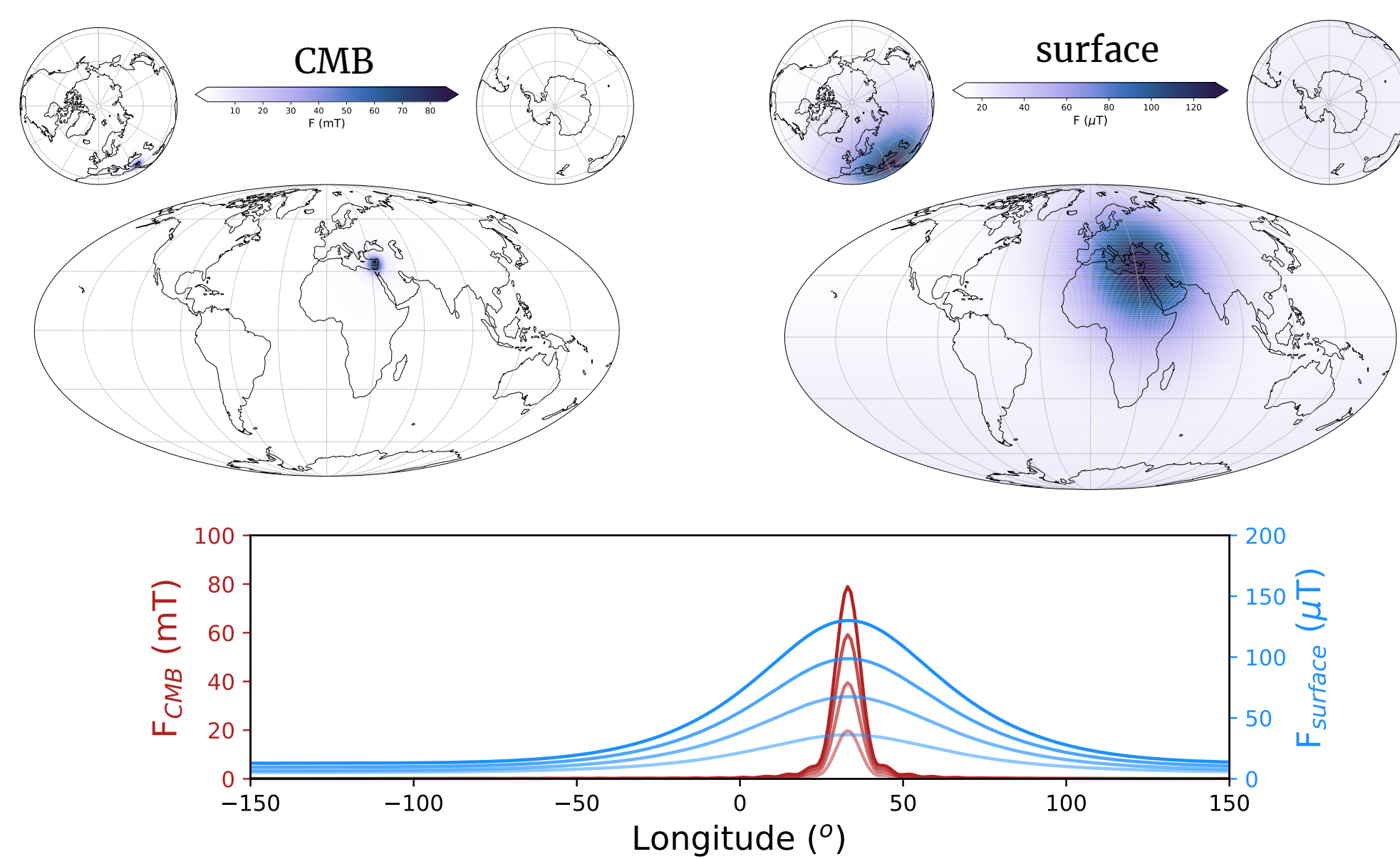


Figure 1: Field intensity at the CMB (top left) and the surface (top right) with a dipolar field ($g_{10} = -26 \mu\text{T}$) and a spike ($-25, -50, -75, -100 \text{ mT}$) at 32°N and 35°E . Cross section (bottom) at 32°N to show spike structure.

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Observations

Toy model of Levantine Iron Age Anomaly

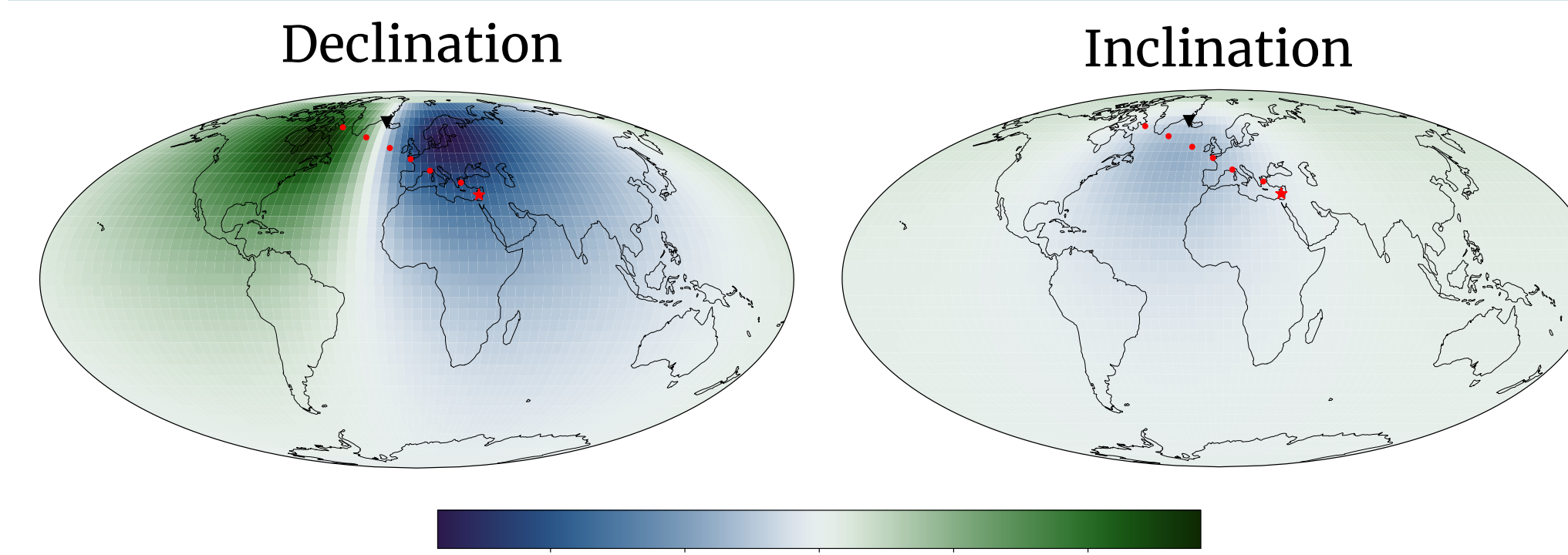


Figure 2: GREENICE15k (black triangle) sampling at the CMB based on directional changes from GAD. Red circles are the LIAA spikes migrating northwest.

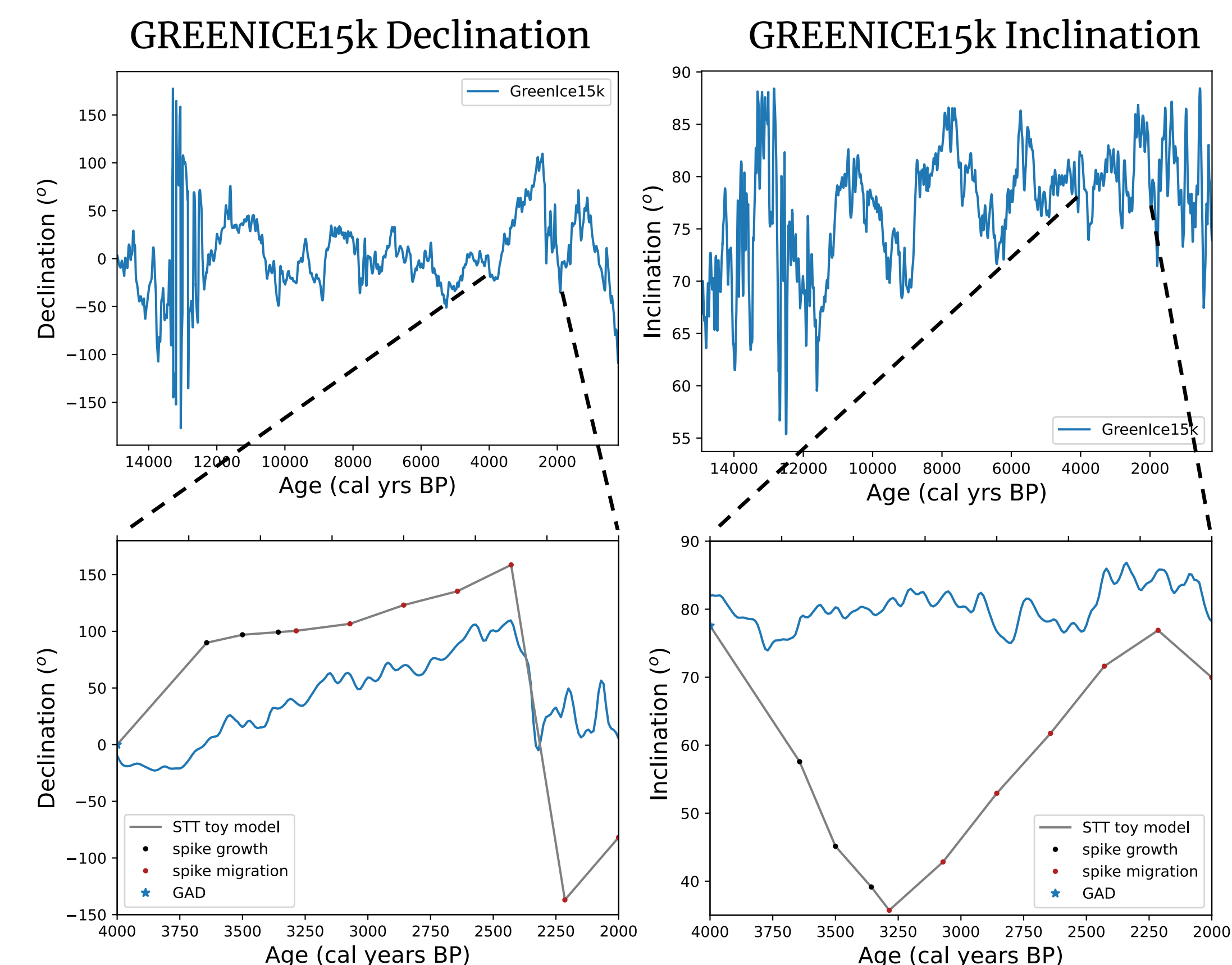
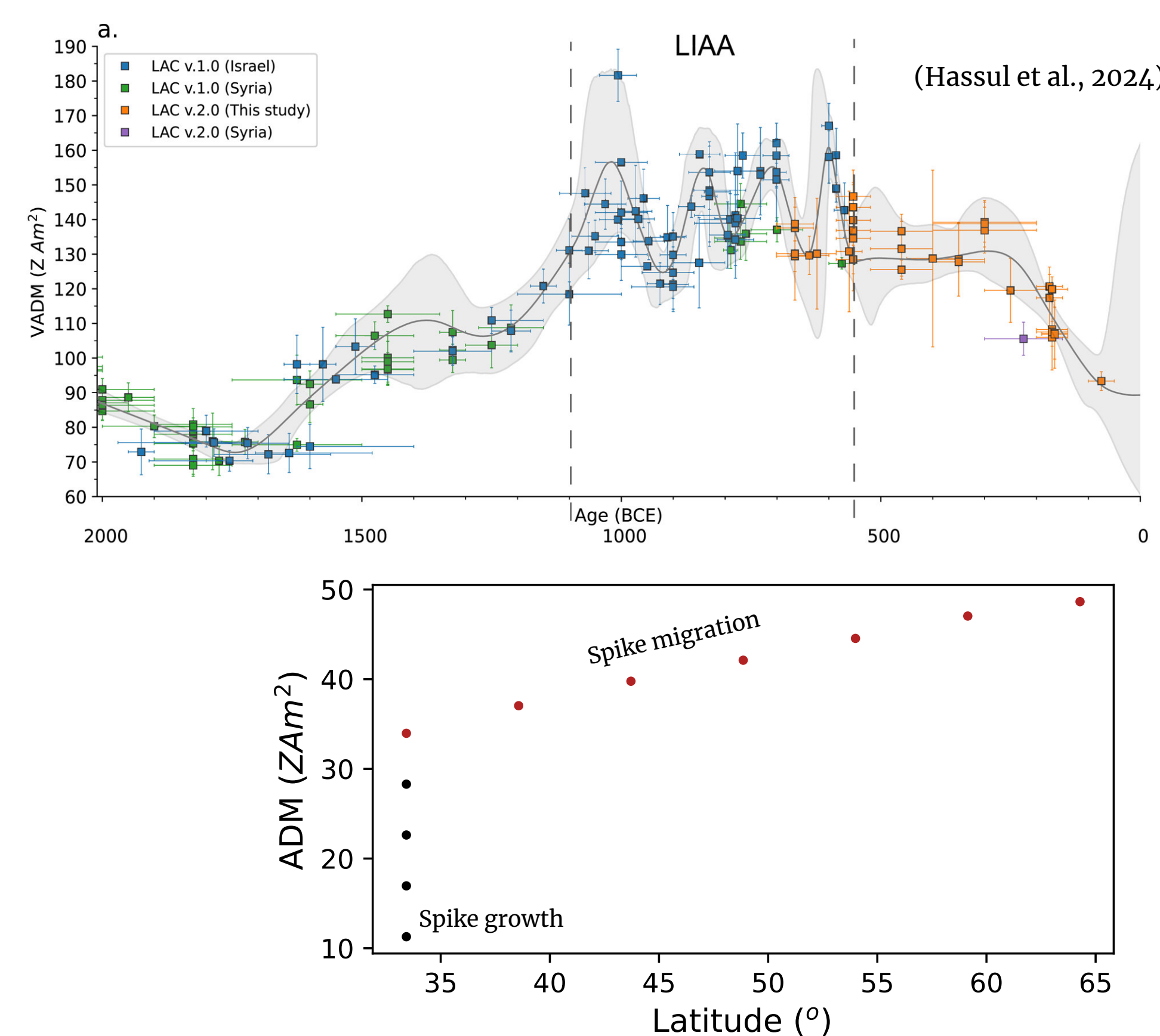


Figure 3: Top row: Declination and inclination of the high resolution paleomagnetic record GREENICE15k (Reilly et al., 2023). Bottom row: Declination and inclination predictions of spike growth (black circles) and migrating NW (red circles). The predictions are for the GREENICE15k location.

Figure 4: Axial dipole moment as a function of spike's latitude. Spike growth (black circles) and migration (red circles).

Toy model of planetary-scale gyre

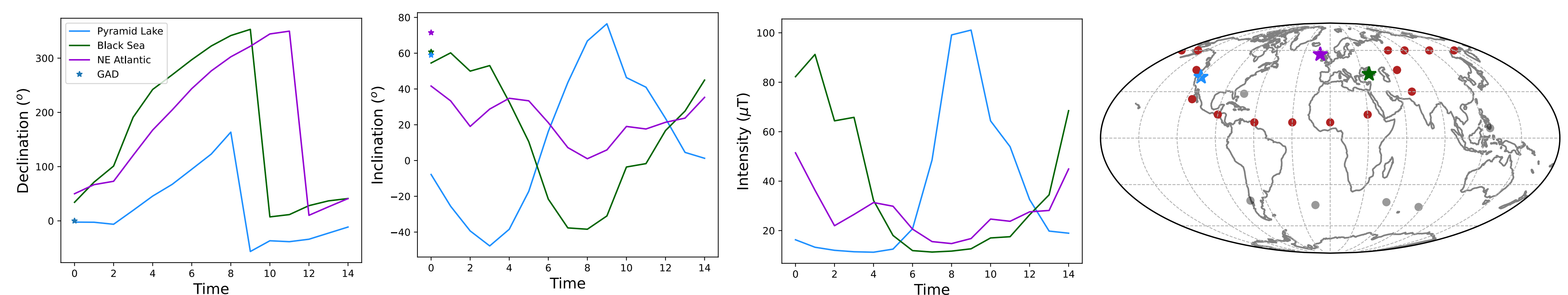


Figure 5: Declination, inclination, and intensity predictions of an intense flux patch migrating (red circles in map) in an anti-cyclonic motion in the northern hemisphere. Predictions are for Pyramid Lake (blue), Black Sea (Green), and NE Atlantic (purple).

Conclusions

- Growth of the LIAA spike at 32°N , depresses the inclination at GreenIce15k (67°N)
- Migration of spike to the northwest allows gradual recovery to the GAD inclination direction
- Presence of LIAA has a large impact on declination at GreenIce15k
- Declination is more sensitive at higher latitudes compared to inclination

References

- Finlay, C.C., Aubert, J., Gillet, N., 2016. Gyre-driven decay of the Earth's magnetic dipole. *Nat. Commun.* 7, 1–8.
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Corresponding author: nclizzie@ucsd.edu